

Symbol creation

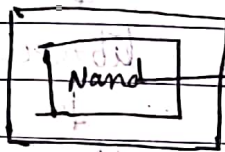
create → cellview from cellview
Symbol generation options

Left Pins → Vin Vb

right Pins → Vout

Top → Vdd

bottom → GND.



edit it.

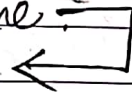
check and save.

* Test bench

go to file



New → select library name
cell name.tb.



go to create → instance

Browse



library Name search



cell → nand.



Hide

And Place the Symbol.



Now go to Create → Instance → browse

$$\begin{aligned} V_1 &= 0 \\ V_2 &= 1 \end{aligned}$$

library Name

cell → Vpulse

view → Symbol

Now go to create → instance → browse

$$\begin{aligned} V_1 &= 0 \\ V_2 &= 0 \end{aligned}$$

library name

cell → Vpulse

view → Symbol

Now go to create → instance → browse

DC Voltage → 1

library name

cell → vdc

view → Symbol

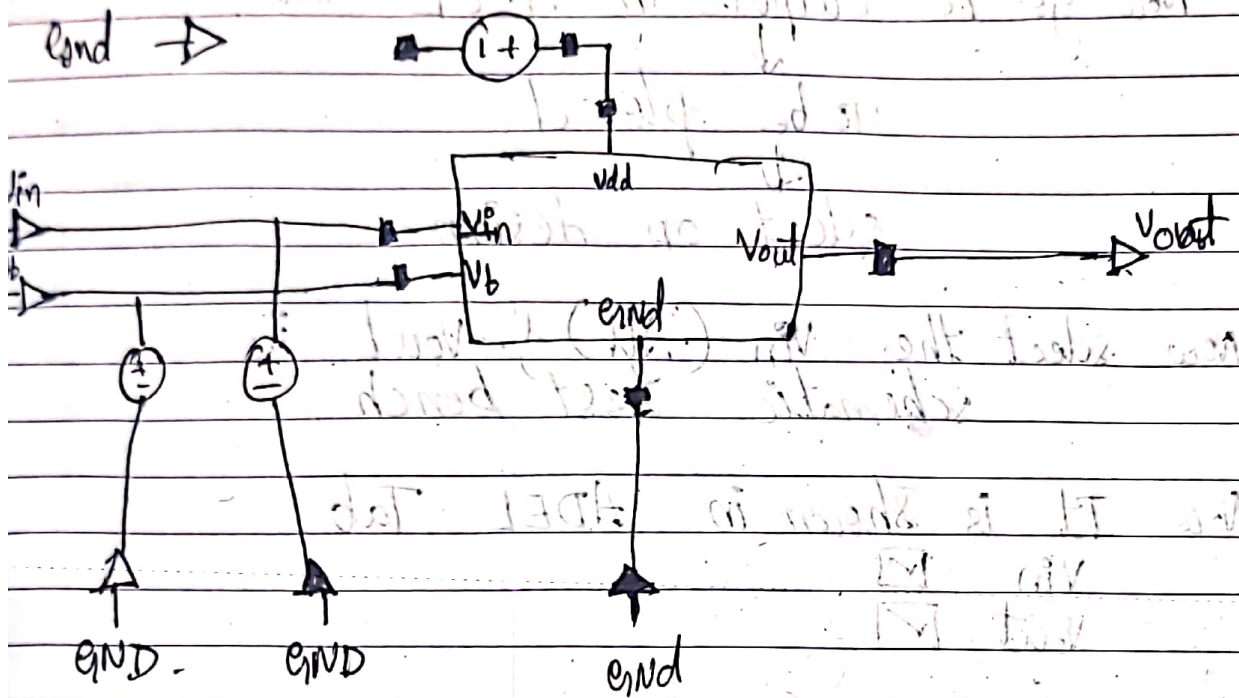
create Pin → gnd & gnd & gnd & gnd.

create Pin → Vin & Vb

create Pin → Vout

~~create pin →~~

And Now connect



lanch ADEL

Analysis → choose → DC → Component Parameter select

DC → start = 0 stop = 1 → select Component from design

component name → OK

Choosing Analysis ADEL

Transient

100n seconds

Moderate ☒

OK

It shows in Analysis

choose Analysis

Now go to outputs in ADEL Tab

↓
To be plotted

↓
select on design

Now select the V_{in} (All) & V_{out} in schematic test bench

Now It is shown in ADEL Tab

V_{in} ☒
 V_{out} ☒

Now click on  green button on Right side the ADEL Tab

Now Waveform is displayed on screen

Truth Table of Nand Gate

Extraction

First open layout

↓
DRC and LVS Run from Assura option.

↓
If No Error (No mismatch).

↓
Again go to Assura of RC Extraction

↓
click on Run Quantus

↓
Left above select setup.

↓
Technology → gpdk180

Rule set → default

Output → Extracted view

Left above select Extraction.

↓
Extraction Type → RC

Ref. Node → gnd!

Cap coupling mode → Coupled.

Enable HRCX → Disable.

At above select Filtering

↓
Enter Power Nets → vdd!
vss!

Enter Ground Nets → gnd!

At above select Netlisting

↓
All do Not include.

And click on **OK**

Now go to Virtuoso Window.

↓
select Tools

↓
Library Manager

↓
give library Name, Ex: (verum)

↓
Cell → Inverter

↓
View → ~~av~~ extracted
(double click):

Now R extracted ← page opens.

↓
zoom and see the Registers & capacitors